

Owen Garnier

Post-doctoral researcher in
Mathematics

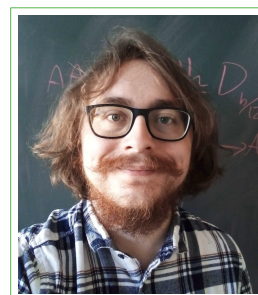
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Studies

- 2024 **PhD in mathematics**, LAMFA, Amiens.
Titre : Garside groupoids and complex braid groups.
Director : Ivan Marin
Other members of the Jury : Yves Guiraud, Jon McCammond, Adrien Brochier, Cédric Bonnafé, David Chataur, François Digne, Eddy Godelle.
- 2018–2021 **Masters degree in mathematics (Algebra, number theory and applications)**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).
Thesis: Théorie de Garside, applications aux groupes de tresses complexes. Cas de G_{31} . (supervisor: Ivan Marin).
- 2020 **Agrégation externe de mathématiques**, Rank: 10/387.
- 2015–2018 **Licence in mathematics**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).

Professional experiences

- 2025–2026 **Post-doctoral researcher**, University of Sevilla, supervised by María Cumplido Cabello.
- 2024–2025 **Temporary research and teaching assistant (ATER)**, Université de Picardie Jules Verne, Amiens.

Distinctions and prizes

- 2025 **PhD thesis award of the Université de Picardie Jules Verne**, Award given to the two best PhD thesis in science, technology and health subjects defended in 2024

Research papers

- 2024 **Uniqueness of roots up to conjugacy in circular and hosohedral-type Garside groups.**
Journal of Group Theory 27, 5 (2024), pp. 1091–1128.
- 2024 **Generalization of the Dehornoy-Lafont order complex to categories. Application to exceptional braid groups.**
Applied Categorical Structures 32, 1 (2024) Paper No. 1, 28 pp.
- 2023 **Regular theory in complex braid groups.**
Journal of Algebra 620 (2023), pp. 534–557.

Preprints

- 2025 **Conjugacy invariants and rigidity in Garside groups: a uniformity phenomenon,**
With M. Calvez, J. González-Meneses and B. Wiest

- 2025 **Normalisers of parabolic subgroups of Artin-Tits groups and Tits cone intersections,** *With E. Heng, A. Licata and O. Yacobi*
- 2025 **Proof of Shvartsman's conjecture on braid groups of projective complex reflection groups.**
- 2024 **Parabolic subgroups of complex braid groups: the remaining case.**
- 2023 **Springer categories for regular centralizers in well-generated complex braid groups.**

Communications and Talks

Talks in conferences

- 2025 **Sous-groupes paraboliques des trickle groups,** *Workshop on trickle groups and their applications, Tours.*
- Système de réécritures et solutions au problème du mot,** *Workshop on trickle groups and their applications, Tours.*
- Sous-groupes paraboliques des groupes de tresses complexes,** *Journées jeunes chercheurs et chercheuses en topologie, Amiens.*
- Groupeïdes de Garside et groupes de tresses complexes,** *Colloque tournant du RT algèbre, Clermont-Ferrand.*
- 2023 **Homology computations for complex braid groups with the Dehornoy-Lafont order complex,** *2023 Meeting of the GDR Homotopy theory and applications, Lille.*
- Bessis braid category for the complex braid group B_{31} ,** *Winterbraids XII, Tours.*
- 2022 **Regular theory in complex braid groups,** *Journées tresses 2022, Amiens.*

Talks in seminars

- 2025 **Normalizers of parabolic subgroups in Artin and Coxeter groups,** *Group theory seminar of the ICMAT, Madrid (Spain)*
- Projective complex braid groups,** *Algebra and Topology Seminar, Canberra (Australia).*
- Sous-groupes paraboliques, des tresses aux groupeïdes de Garside,** *Séminaire Géométrie, Algèbre, Dynamique et Topologie, Dijon*
- Groupeïdes de Garside paraboliques et applications aux groupes de tresses complexes,** *Séminaire d'algèbre et de géométrie, Caen*
- 2024 **Des partitions non-croisées à l'étude des sous-groupes paraboliques des groupes de tresses complexes,** *Séminaire SPACE, Tours*
- Sous-groupes paraboliques et éléments réguliers des groupes de tresses complexes,** *Gaston Darboux seminar of the IMAG, Montpellier*
- Homology of categories and the Dehornoy-Lafont order complex,** *Seminar of the "Ouragan" team, Paris*
- 2024 **Description de l'espace des orbites régulières d'un groupe de réflexions complexes,** *Topology seminar of the laboratoire Paul Painlevé, Lille*
- 2023 **Étude Garside du groupe de tresses complexe $B(G_{31})$,** *GATo Seminar of the LAMFA, Amiens*
- Garside study of the complex braid group $B(G_{31})$,** *Seminar "Groupes, Représentations et Géométrie" of the IMJ-PRG, Paris*
- Homology of a category and the Dehornoy-Lafont order complex,** *Paris algebra seminar, Paris*

- 2022 **Regular theory in complex braid groups**, *Seminario del departamento de Álgebra*, Sevilla (Spain).
Théorie régulière dans les groupes de tresses complexes, *GATo seminar of the LAMFA*, Amiens.

Conferences

- 2025 **Workshop on trickle groups and their applications**, Tours, from 25/06 to 27/06.
Artin groups, Braids and mapping class groups: Celebrating the work of Luis Paris, Cáceres (Spain), from 16/06 to 20/06 .
Journées jeunes chercheurs et chercheuses en topologie, Amiens, from 09/04 to 11/04.
Colloque tournant du RT algèbre, Clermont-Ferrand, from 03/05 to 03/07.
Winterbraids XIV, Bordeaux, from 02/03 to 02/06.
- 2023 **2023 Meeting of the GDR Homotopy theory and applications**, Lille, from 10/24 to 10/27.
Various Guises of Reflection Arrangements workshop, Edimbourg (UK), from 03/13 to 03/17.
Spring School on Real, Complex, and Symplectic Reflection Groups, Bochum (Germany), from 03/06 to 03/10.
Winterbraids XII, Tours, from 02/21 to 02/24.
- 2022 **Journées tresses 2022 : Groupes de tresses généralisés**, Amiens, from 08/29 to 09/02.
- 2021 **Summer school: The dual approach to Coxeter and Artin groups; Garside theory and applications**, Berlin (Germany), from 08/30 to 09/04.

Teaching

- 2024–2025 **Exercices courses**, *Université de Picardie Jules Verne*, Amiens.
- Calculus (first year) (30h).
 - Set theory (second year) (20h).
 - Groups, Rings, Fields (third year) (36h).
 - Linear algebra 1 (first year) (32h).
 - parametrized curves (first year) (36h).
 - Complex analysis (third year) (20h).
- 2023–2024 **Exercices courses**, *Université de Picardie Jules Verne*, Amiens.
- Advanced linear algebra (third year) (20h).
 - Complex analysis (third year) (20h).
- 2022–2023 **Exercices courses**, *Université de Picardie Jules Verne*, Amiens.
- Advanced linear algebra (third year) (20h).
 - Algebraic algorithmics (third year) (20h).
- 2021–2022 **Exercices courses**, *Université de Picardie Jules Verne*, Amiens.
- Calculus (first year) (24h).
 - Advanced linear algebra (third year) (20h).
 - Complex numbers and geometry (third year) (20h).

Other activities

- 2025 **Co-organizer of the workshop on trickle groups and their applications**, Tours.
With Igor Haladjian.

- 2023–2024 **Co-organizer of the PhD student seminar of the LAMFA**, Amiens.
With Nicolas Beuvin et Michaël Schoonheere.
- 2022–2023 **Co-organizer of the PhD student seminar of the LAMFA**, Amiens.
With Felipe Arbulú et Ismaïl Razack.
- 2022 **Research stay in Sevilla**, *Universidad de Sevilla*, Sevilla (Spain).
Work with Juan Gonzalez-Meneses (3 weeks).
- 2022 **Co-organizer of the braid meeting 2022**, Amiens.
With Ivan Marin and Karine Sorlin.